

03 DFS

Lab 2b: Depth-First Search

Same lab, same query: A to H. Now change one line of your BFS program to make it depth-first.

Answer this: DFS visits fewer nodes here than BFS. Is DFS therefore the better algorithm? Move the goal from under E to under G and re-run before answer.

⇒ what is DFS? —

DFS stands for Depth-First-Search

"It will go as deep as possible in one path first. If it find a dead end, it will come back and try another path."

So DFS : — i) choose one path

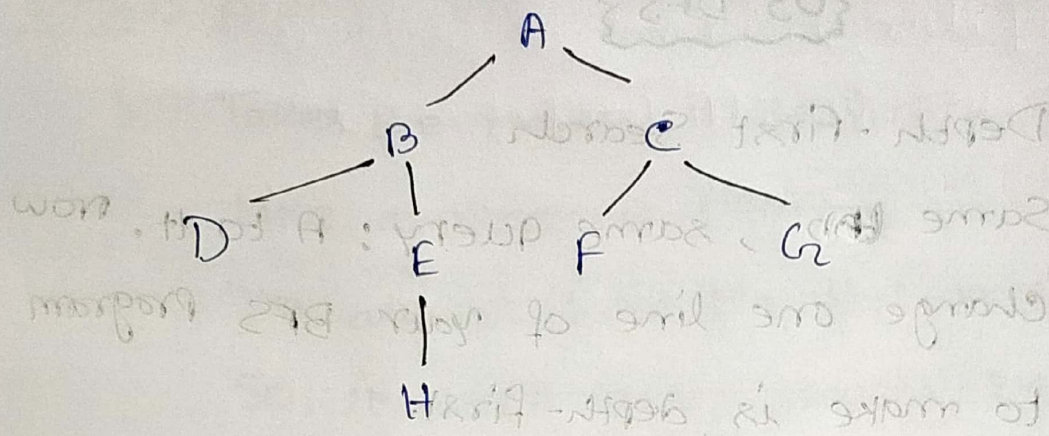
ii) goes deeper and deeper

iii) It stuck, comes back

(Backtracking).

iv) Tries another path

• Tree : —



we want to go from $A \rightarrow H$

● Step-by-Step DFS

i) Initial Stack $\Rightarrow [A]$

ii) Step 1 $\Rightarrow$ POP A

visited : A

Push C then B (So B is visited first)

Stack : [C, B]

iii) Step 2 $\Rightarrow$ POP B

visited : A, B

Push E then D

Stack : [C, E, D]

iv) Step 3 $\Rightarrow$ POP D

visited : A, B, D

D has no children

Stack : [C, E]

v) Step 4 $\Rightarrow$ POP E

visited : A, B, D, E

Push H

stack: [C, H]

vi) Step 5 $\Rightarrow$ POP H

visited: A, B, D, E, H

Goal Found!

Path: A $\rightarrow$ B $\rightarrow$ E $\rightarrow$ H

Python Program :-

1. graph = {

'A': ['B', 'C'],

'B': ['D', 'E'],

'C': ['F', 'G'],

'D': [],

'E': ['H'],

'F': [],

'G': [],

'H': []

}

stack = [('A', ['A'])]

while stack:

node, path = stack.pop()

if node == 'H':

print("path:", path)

break

for child in reversed(graph[node]):

~~Stack~~ stack.append((child, path + [child]))

H ← E ← D ← B ← A : path

! must loop

H ← E ← D ← B ← A : path

Path from Problem :

graph = {

A : [B, C]

B : [D, E]

C : [F, G]

D : [H]

E : [I]

F : [J]

G : [K]

DFS visits fewer nodes here than BFS, is DFS therefore the better algorithm? —

i) no

ii) DFS is not always better

iii) In the first tree, DFS found H quickly and visited fewer nodes

iv) But after moving H under G, DFS had to search many more nodes before finding the goal

v) The performance of DFS depends on where the goal is located.

vi) Sometime DFS is faster, sometimes BFS is better

Therefore, DFS is not always the better algorithm